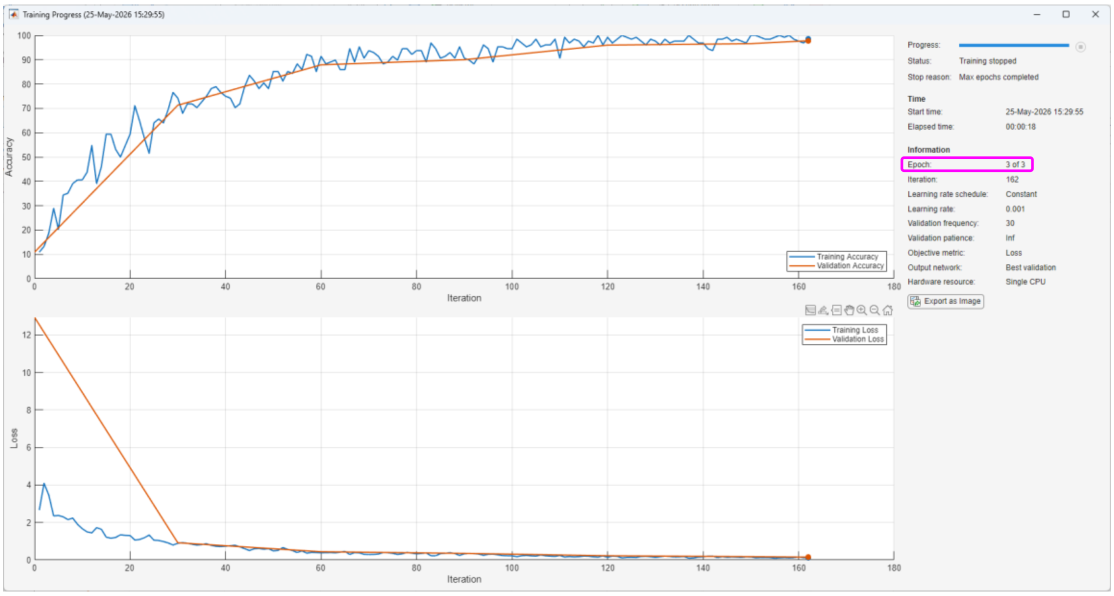
1. Modify the code of "CreateSimpleDeepLearningClassificationNetworkExample.mlx" according to the following statements and run the modified code.
   a) Split the data into 70% training and 10% validation, 20% test data.
   b) Change the solver "sgdm" into "adam," and name-value pair "MaxEpochs=3."
   c) Use the "ValidationData=imdsValidation," but calculate accuracy = testnet(net,imdsTest,"accuracy").

   Example Output:



   My result as accuracy = 96.4500

   The answer of EX1: submit the file (.mlx) to the system.

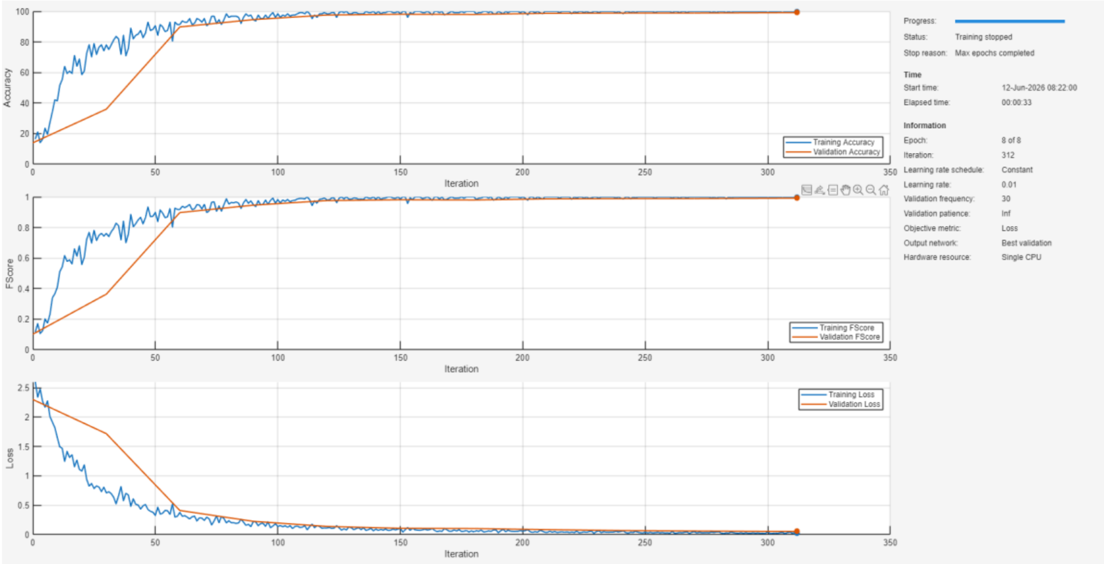
2. Modify the code of Modify the code of "PlotAccuracyDuringTrainingExample.mlx" according to the following statements and run the modified code.

   Add a group of four layers into the original layers (14 layers) and modify "layers" to make the final layer graph as in Figure 2-2. This group of code **looks like** the code below:

```
convolution2dLayer(3,32,Padding="same")
batchNormalizationLayer
reluLayer
maxPooling2dLayer(2,Stride=2)
```

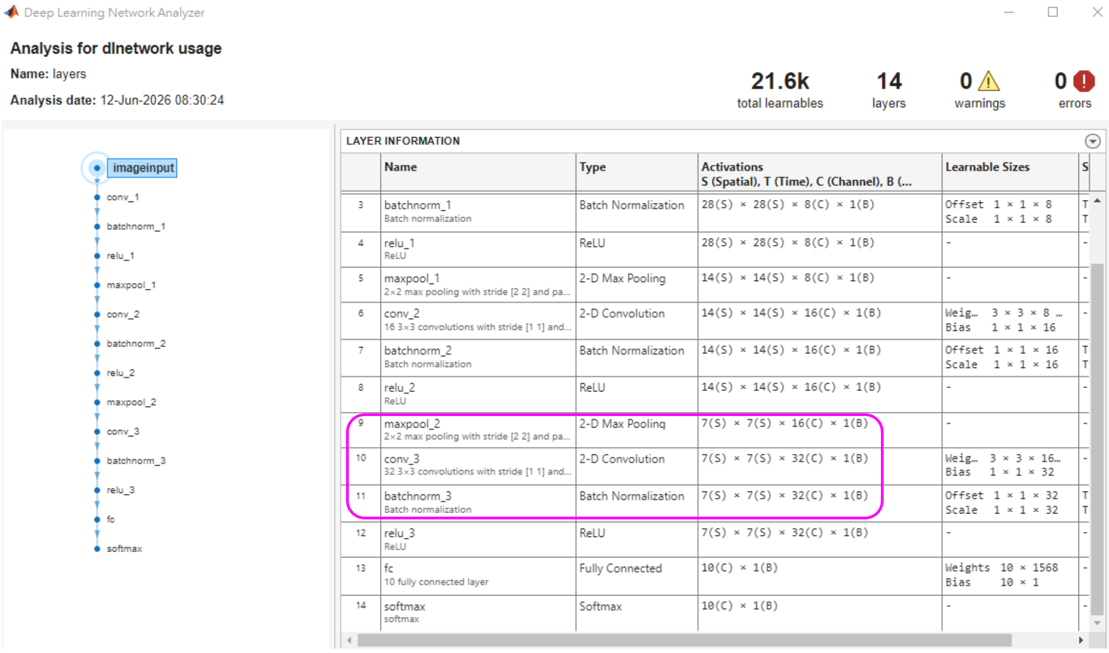
   Use analyzeNetwork(layers) to see the details:

   Original results:



analyzeNetwork(layers)

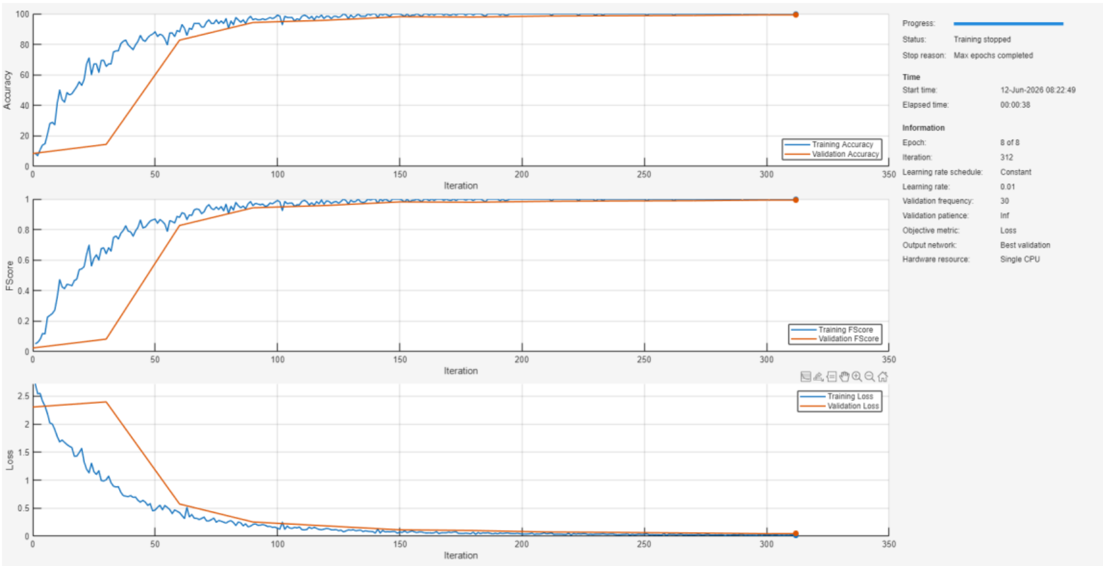
The number of the original layers is 14.



	Name	Type	Activations S (Spatial), T (Time), C (Channel), B (...	Learnable Sizes	S
3	batchnorm_1 Batch normalization	Batch Normalization	28(S) × 28(S) × 8(C) × 1(B)	Offset 1 × 1 × 8 Scale 1 × 1 × 8	T T
4	relu_1 ReLU	ReLU	28(S) × 28(S) × 8(C) × 1(B)	-	-
5	maxpool_1 2×2 max pooling with stride [2 2] and pa...	2-D Max Pooling	14(S) × 14(S) × 8(C) × 1(B)	-	-
6	conv_2 16 3×3 convolutions with stride [1 1] and...	2-D Convolution	14(S) × 14(S) × 16(C) × 1(B)	Weig... 3 × 3 × 8 ... Bias 1 × 1 × 16	-
7	batchnorm_2 Batch normalization	Batch Normalization	14(S) × 14(S) × 16(C) × 1(B)	Offset 1 × 1 × 16 Scale 1 × 1 × 16	T T
8	relu_2 ReLU	ReLU	14(S) × 14(S) × 16(C) × 1(B)	-	-
9	maxpool_2 2×2 max pooling with stride [2 2] and pa...	2-D Max Pooling	7(S) × 7(S) × 16(C) × 1(B)	-	-
10	conv_3 32 3×3 convolutions with stride [1 1] and...	2-D Convolution	7(S) × 7(S) × 32(C) × 1(B)	Weig... 3 × 3 × 16... Bias 1 × 1 × 32	-
11	batchnorm_3 Batch normalization	Batch Normalization	7(S) × 7(S) × 32(C) × 1(B)	Offset 1 × 1 × 32 Scale 1 × 1 × 32	T T
12	relu_3 ReLU	ReLU	7(S) × 7(S) × 32(C) × 1(B)	-	-
13	fc 10 fully connected layer	Fully Connected	10(C) × 1(B)	Weights 10 × 1568 Bias 10 × 1	-
14	softmax softmax	Softmax	10(C) × 1(B)	-	-

Figure 2-1.

Modified results:



analyzeNetwork(layers)

The number of the modified layers is 18 (14 + 4).

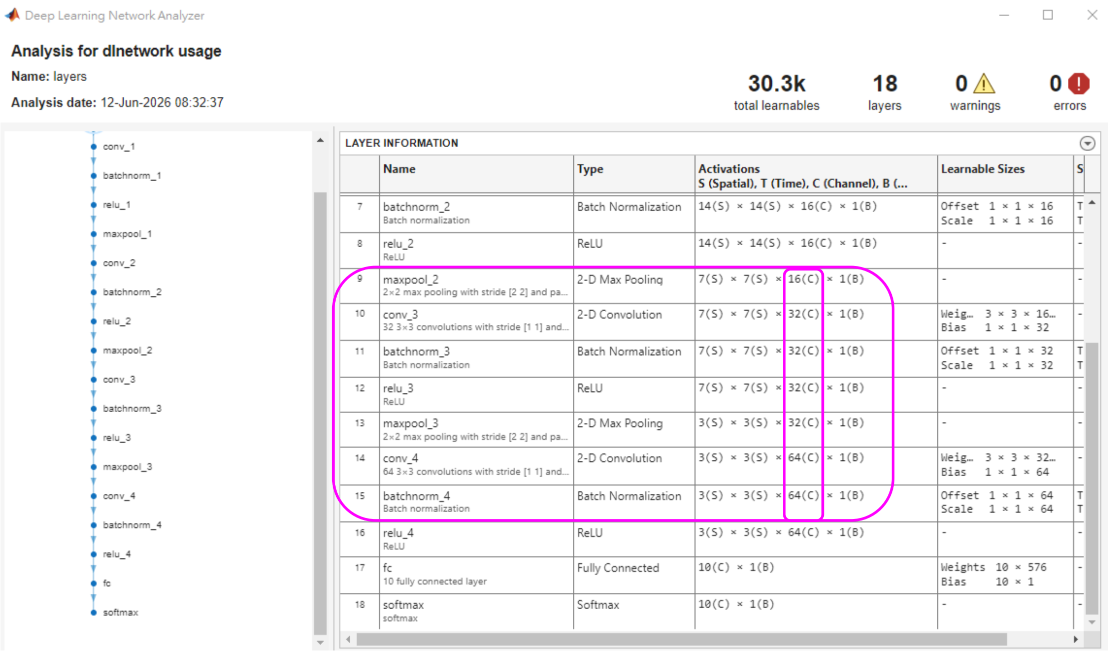
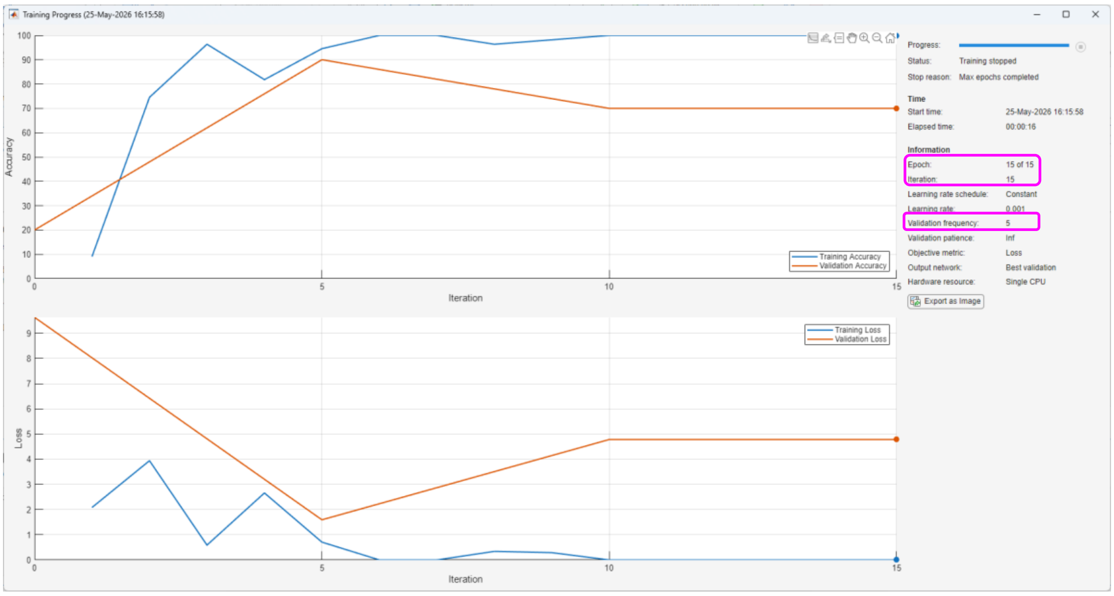


Figure 2-2.

The answer of EX2: submit the file (.mlx) to the system.

3. Modify the code of "CreateSimpleDeepLearningClassificationNetworkExample.mlx" to run for "MerchData," not the original "DigitsData" given inputSize = [227 227 3]. In the trainingOptions, "Solver" as "adam," "Epoch" as "15" and "Validation frequency" as "5." In addition, "imdsTrain" occupies 70%, "imdsValidation" 15%, "imdsTest" 15%.

Example Output:



My result is as follows:

```
accuracy = testnet(net,imdsTest,"accuracy")
```

```
accuracy = 80
```

The answer of EX3: submit the file (.mlx) to the system.

4. Modify the code of "ClassifyImageUsingPretrainedNetworkExample.mlx" according to the following statements and run the modified code.

a) Change pretrained model "googlenet" as "squeezenet."

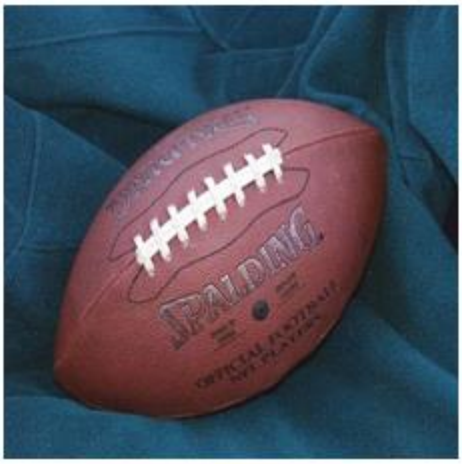
b) The original picture "peppers.png" is replaced as "llama.jpg," "football.jpg," and "sherlock.jpg" one by one. Therefore, you need to expand some codes to present three recognized pictures simultaneously on the file.

Example Output:

Picture: llama.jpg

Picture: football.jpg

buckle



Picture: sherlock.jpg

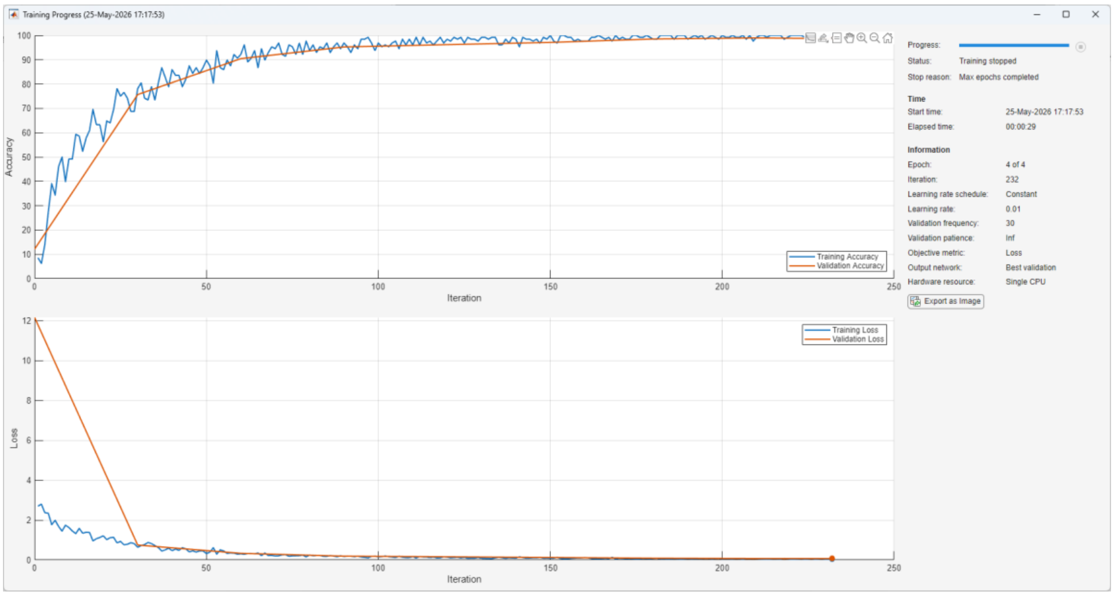
golden retriever

The answer of EX4: submit the file (.mlx) to the system.

5. Modify the code of "CreateSimpleDeepLearningClassificationNetworkExample.mlx" according to the following statement and run the modified code.

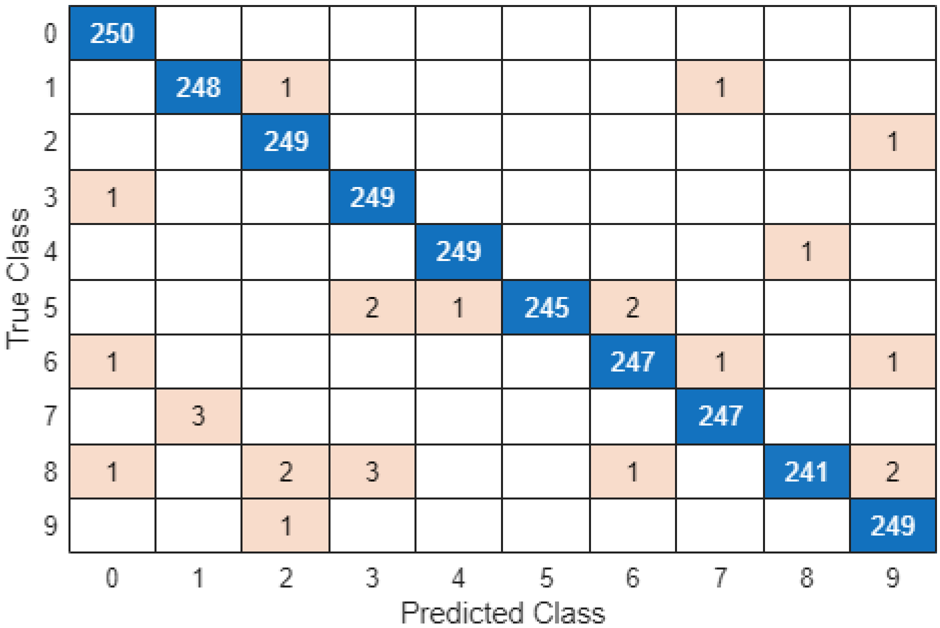
a) Add part of code from "GetStartedWithTransferLearningExample.mlx" into the previous file to present the confusion matrix.

Example Output:




```
accuracy = testnet(net,imdsValidation,"accuracy")
```

```
accuracy = 98.9600
```



The answer of EX5: submit the file (.mlx) to the system.